COMPANY A / CONSULTING PROOF OF CONCEPT

Retain contribution. Not just customers.

Risk ≠ value

A value-aware service-recovery strategy, grounded in 100,000 customer records—and designed to earn its right to scale.

philopateer | GCI World 2026 | Local Python / CPU analysis

Company A is anonymous. Model evidence is observed; intervention economics are explicitly conditional.

Approve evidence—not a blanket discount campaign

SAMPLE BENCHMARK

+42.4%

More churned recurring value captured than risk-only. Sample top 10% of rows; not a deployment forecast or measured savings.

CONDITIONAL ECONOMICS

$3.7k

At 2% assumed churn in a 100,000-customer cohort. Nominal 10% outreach; costs, response and population transport are unverified.

MANAGEMENT DECISION

Pilot first. Earn the rollout.

Test opt-in service recovery against usual care. The pilot buys causal learning—not proof of profitable rollout.

Decision ladder: risk-only → value-aware triage → causal incremental-profit targeting.
Keep the verified LightGBM baseline; scale only after fully loaded contribution and harm gates pass.

[D1, S5–S6] Sample and deployment regimes are separate. Economics are conditional USD-equivalent assumptions. Risk ≠ response.

Retention economics matter more than contact volume

CURRENT MARKET CONTEXT

0.93% / 2.72%

T-Mobile FY2025 postpaid / prepaid [S1].
Monthly rate uses average customers.
Postpaid phone ARPU: $50.37 per month.
AT&T: 0.90%; beginning-month denominator [S2].

6% → 10% churn

A specific randomized plan-recommendation campaign increased three-month churn—not a universal effect. [S4]

COST DISCIPLINE PRECEDENT

30% lower

Verizon reports improved promotional retention cost, Q3 2026 vs prior year.
Management claim—not causal evidence. [S3]

Implication for Company A
Offer useful, permission-based help.
Measure adverse effects and all costs.
Keep an untreated comparison group.

[S1–S4] Carrier segments, denominators and horizons differ. None establishes Company A prevalence or this workflow's treatment effect.

100,000 customers; a useful sample, not a live forecast

100,000

Customers; 100 merged columns.
No unmatched or duplicate IDs.

49.56%

Observed sample churn.
Not a population churn estimate.

Days 31–60

Label window after observation.
Not cumulative 60-day churn.

What we can establish

Behavioral segments, held-out risk ranking and recurring-value prioritization.
133 invalid negative handset ages are handled; billing credits are not blindly deleted.

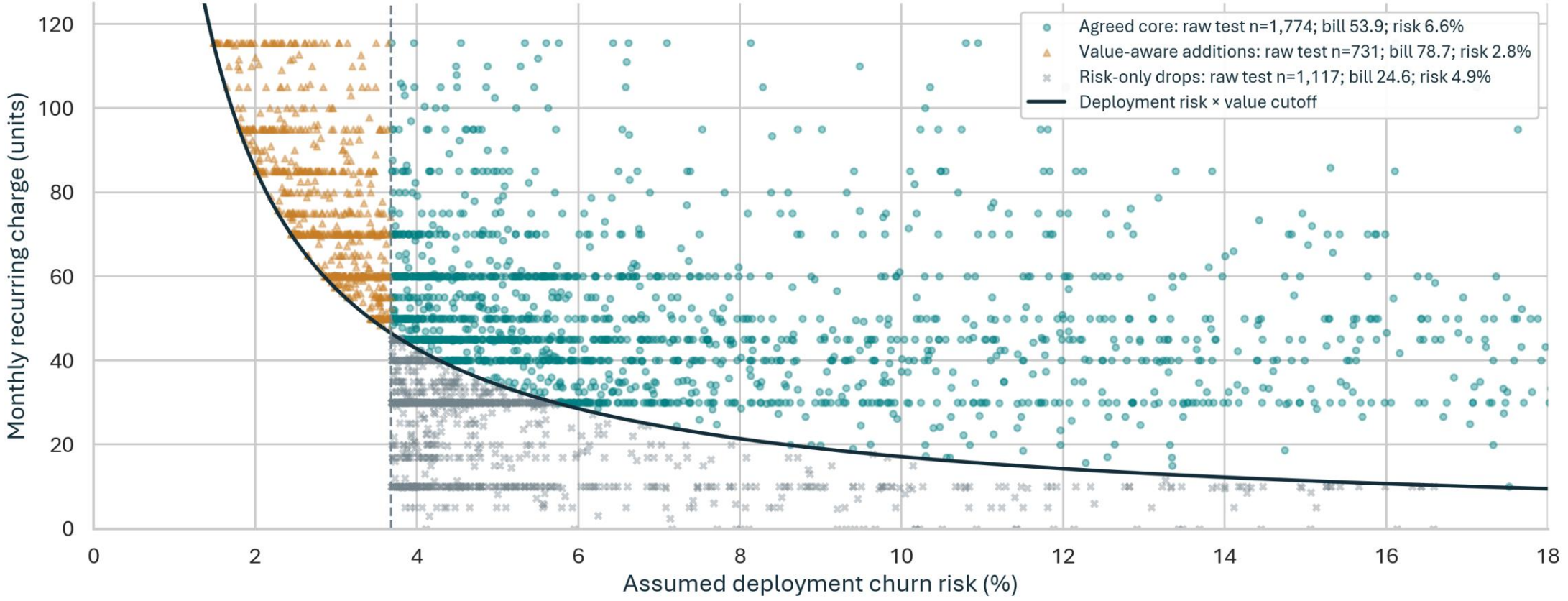
What must be acquired

Current dates, live baseline, true margins, offer costs, household linkage and randomized treatment outcomes.
Currency and sampling design are unverified.

[D1–D2] Original data audit. Customer-ID order varies with tenure and churn; random holdout is not temporal validation. Source files preserved and fingerprinted.

Risk ≠ value: who gets the discretionary call?

Deployment scenario: 2% prior; risk and value-aware cutoffs frozen on calibration rows. Mandatory defect remediation is not rationed.

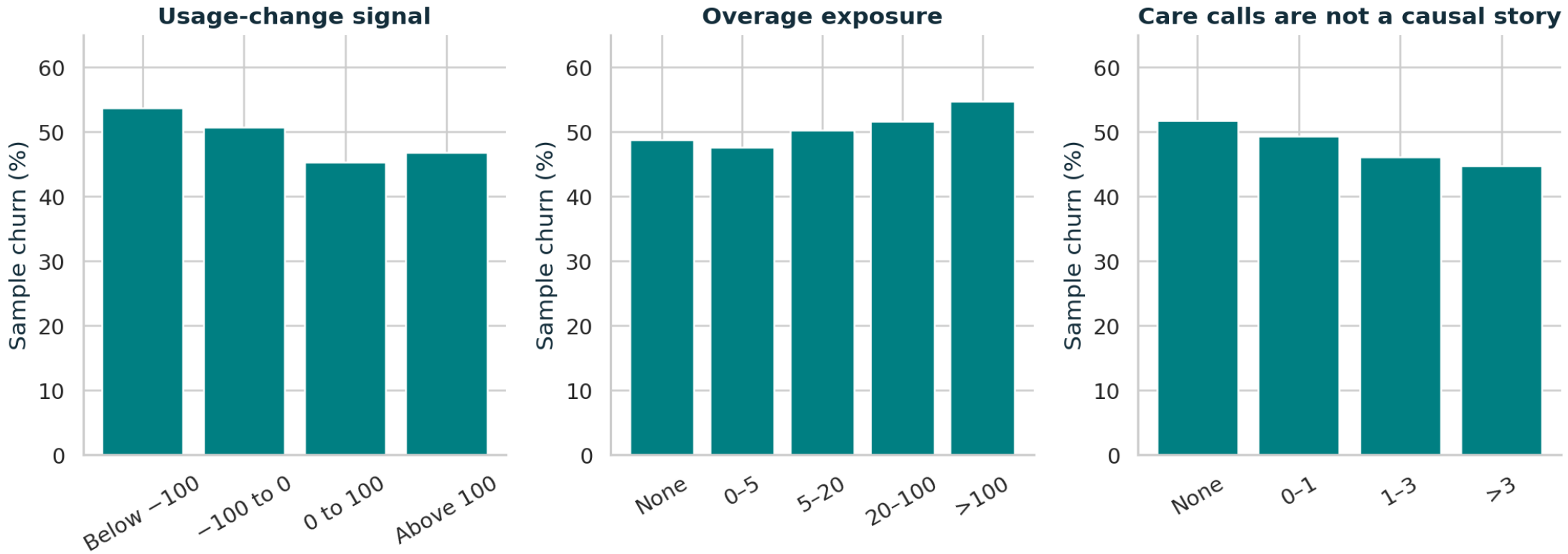


Value-aware triage adds higher-bill accounts and drops lower-bill ones—not a causal claim about who will respond.

[D1, M5–M6] Raw n are sample counts; bill/model-risk means are deployment-weighted. No point subsampling; axes crop the display.

Diagnose the issue before choosing the remedy

Training-sample patterns suggest questions for an agent—not automatic price cuts, device swaps or causal conclusions.



Usage decline → ask what changed. Overage exposure → explain the bill. Care activity → verify whether the issue was resolved.

[D1–D2] Training n=60,000. Usage-change units are ambiguous; raw thresholds are not labeled percentages. Overage values are dataset monetary units.

Separate selection, calibration and final testing

Previously evaluated baseline test reused; enhancement decisions frozen before the descriptive final batch. Original split preserved.

60,000

Train the candidates

15,000

Select by ROC-AUC

10,000

Calibrate; set policy

15,000

Final held-out test

87 operational features

Sensitive fields excluded; operational proxies remain.
Credit-card AUC 0.657; ownership 0.607;
high ordinal income 0.627 (fold-local label).
Missing targets dropped; subgroup audit ≠ fairness.
No Range fields or inferred Mean windows.

Headroom gate: baseline retained

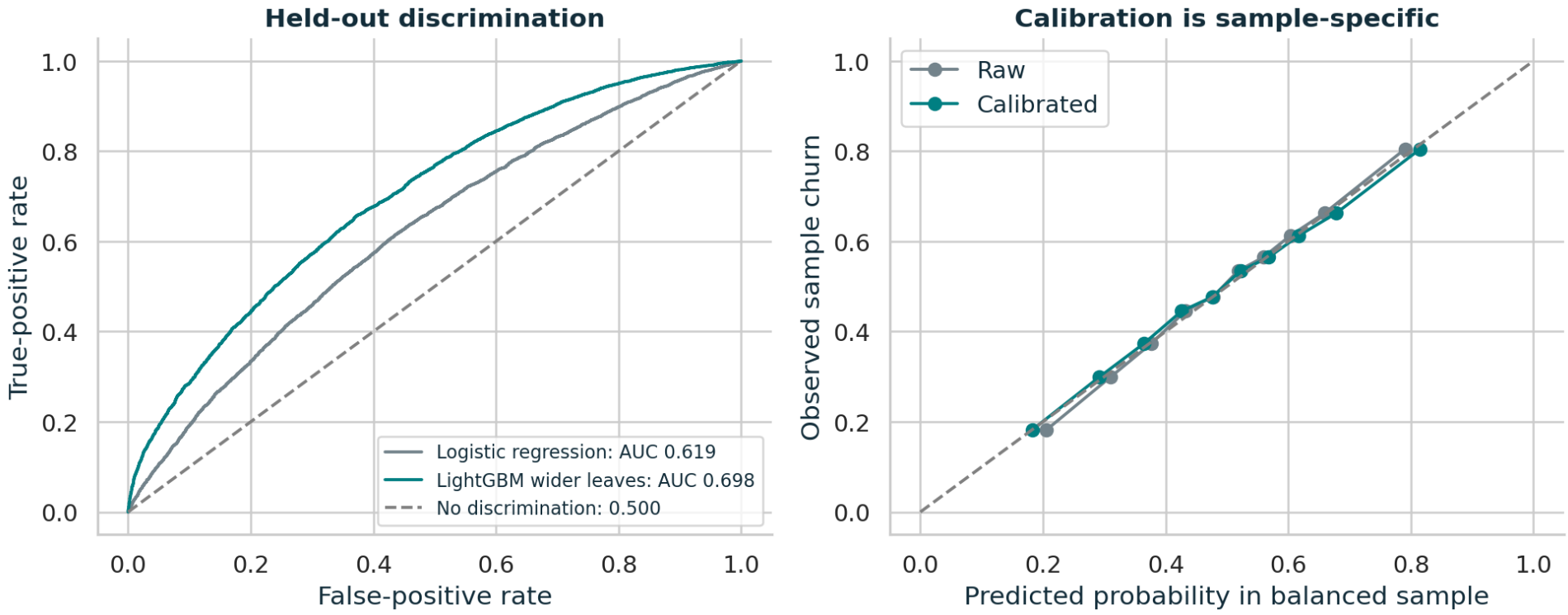
Repeated 2 × 5 non-test CV on 75,000 rows.
Best blend mean ΔAUC: +0.001085.
Corrected 95% CI: −0.000032 to +0.002203.
Tail/convergence gates also failed.
No challenger promoted; no ceiling established.

[D1, M1–M8] Local CPU; seed 20261028. Headroom intervals are exploratory, pointwise—not selection-adjusted. Proxy prediction does not prove independence.

Retained LightGBM: original held-out AUC 0.6978

Original baseline reference; 15,000 reused test customers. No challenger passed the non-test gate; the chart is unchanged.

95% AUC interval: 0.6899–0.7055 **Accuracy: 64.02%** **Brier score: 0.2195**



Ranking signal is useful for triage. Sample-calibrated probabilities are not deployment prevalence or treatment responsiveness.

[D1, M1–M2] Accuracy uses threshold 0.5. AUC interval: 300 test-row bootstrap replicates, conditional on fitted model. Refitting uncertainty not included.

Nominal 10% outreach captures 29.5% of churned value

2% deployment scenario: population-weighted cutoff frozen on calibration. This is not the sample top-decile comparison.



DEPLOYMENT-WEIGHTED VALUE CAPTURE

29.5% Value-aware

15.0% Value only

25.6% Risk only

+15.4% relative to risk only.
95% conditional paired interval:
+11.3% to +19.7%.

Separate sample benchmark: top 1,500 rows capture 20.8% vs 14.6% of churned value; +42.4% relative gain—not retention lift.

[D1, S6, M5] Curve: stable-rank diagnostic; headline: frozen cutoff. Population transport is unverified; conditional intervals exclude refit/selection uncertainty.

One service workflow. A mandatory remediation floor.

Service-Fix Floor — verified defects are never rationed

Remediate every verified billing/network defect regardless of bill size, discretionary selection or campaign arm.
Value-aware ranking only rations scarce proactive outreach; verify consent and linked-account assignment.

Billing check

582 raw rows
68.7 mean bill
23.4% at-risk value

Verify the bill and recent revenue change. Explain and resolve; no inferred promotion expiry.

Network check

266 raw rows
59.7 mean bill
10.4% at-risk value

Inspect failed-call signals. Confirm a live issue before escalation; missing data is not a verified defect.

Handset check

791 raw rows
62.9 mean bill
30.7% at-risk value

Age >365 days flags an opt-in health check.
No automatic replacement or equipment-failure claim.

Digital residual

866 raw rows
64.7 mean bill
35.5% at-risk value

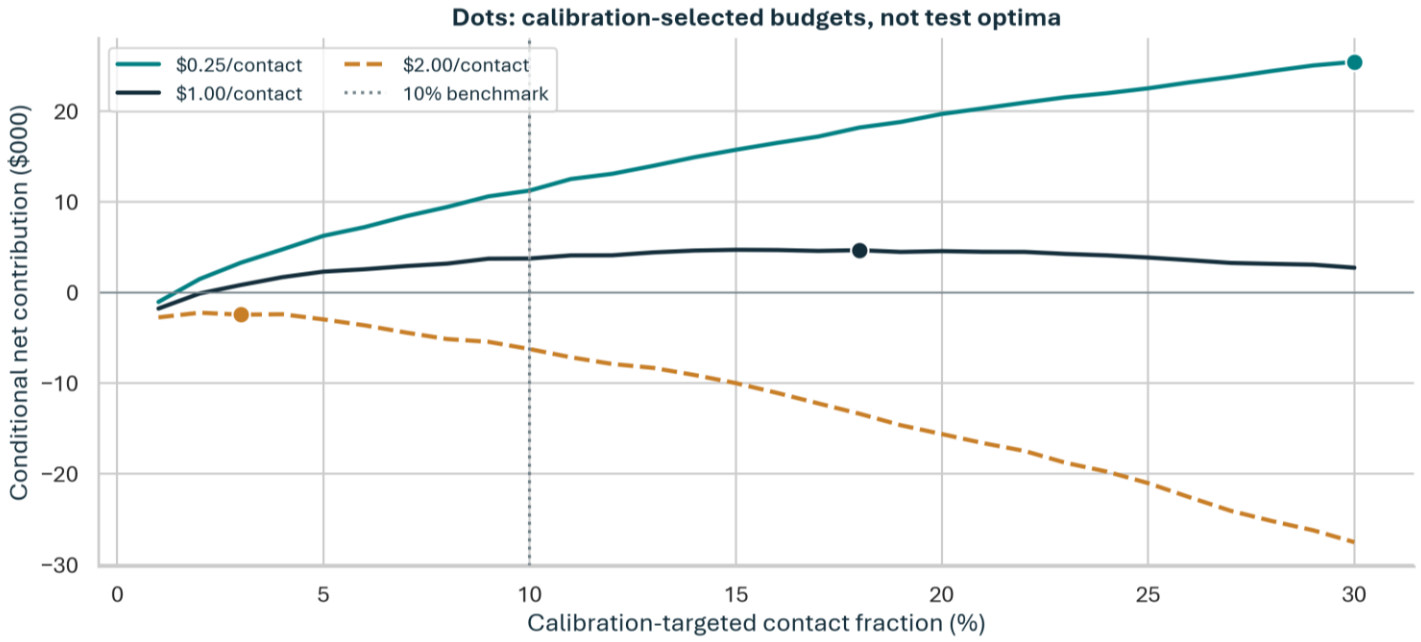
Permission-based digital check for remaining cases. Not uniformly low usage; no proactive billing call.

Exclusive precedence: billing → network → handset → residual. These are action hypotheses, not diagnosed causes.
Sample counts; mean bills and value shares weighted to the 2% scenario. No automatic subsidy or plan switch.

[D1, S4–S8] Frozen calibration routing; descriptive reused-test queues. Comparable diagnostic/agent workflows establish feasibility, not savings.

A conditional $3.7k business case—not a forecast

Hypothetical 100,000-customer cohort; about 9,985 weighted contacts at nominal 10%. Budget unchanged; no test-selected optimum.



ALL ARE ASSUMPTIONS

2% deployment churn

20% of would-be churners

prevented

6 months × 60% margin

$1 per contact; $5k fixed
No blanket subsidy
Net 95% interval: $2.8k–$4.7k

Conditional scenario: 95 prevented → $31.2k retained revenue → $18.7k gross contribution.
No-contact alternative: zero without campaign launch.

[D1, S6, M5] Dots are calibration-selected budgets, not test optima. Costs and benefits are population-weighted; transport/response are unverified. USD-equivalent, not actual currency.

Small cost changes can erase the entire return

Net contribution in USD-equivalent thousands; 2% assumed churn, 100,000-customer cohort, $5,000 fixed cost.

	$1 / contact	$2 / contact	$5 / contact
10% prevented	−5.6	−15.6	−45.6
20% prevented	**+3.7**	−6.2	−36.2
30% prevented	+13.1	+3.1	−26.8

Break-even cost
$1.38 / contact

Break-even prevention
16.0%

Prior stress, same base costs:
1% → −5.6k
2% → +3.7k
5% → +30.7k

No-contact means zero without campaign launch; an empty launched list still incurs fixed cost.
Verify live prevalence, fully loaded support costs and causal effectiveness before committing funds.

[D1, M5–M6] Editable scenario sensitivity, not a confidence interval. Prevention is assumed; break-even holds all other base assumptions fixed.

Fund causal learning—not an assumed profitable rollout.

Two disjoint learning stages

Core: 7,872 linked accounts/arm in both + rv_only.
$12,872 contact/startup envelope.
Expansion: risk_only 2,000/arm; exploration 1,000/arm.
$3,000 incremental envelope; not powered contrasts.
Core needs 15,744 unique accounts across larger portfolios/cohorts.
No duplicated rv_only enrollment; acquire household linkage.
Costs exclude unquoted consent, linkage, follow-up and engineering.

Binary learning ≠ profit proof

Calibration simulation (planning proxy): adjusted contribution needs 80,000/arm; unadjusted not 80% powered through 160,000/arm. No validated longitudinal CUPED or household-clustered power. Re-estimate using live data.

Randomize linked clusters 50/50; log joint enrollment × assignment propensities, costs and outcomes.
HT/IPW Bayes-minus-risk: only both cancels; weight incremental value in rv_only minus risk_only by deployment mass.
Exploration supports below-cutoff learning; no unbiased claim for unconsented or unobserved accounts.

Scale: cluster-robust 95% lower bound of fully loaded incremental contribution >0 and no harm.
Extend: interval overlaps zero, positive upper bound, no harm, prospectively approved sample/budget cap.
Stop: upper bound ≤0, infeasible costs or harm. Serious consent/complaint events escalate independently.

[D1, S4–S6, M7] Binary plan: two-sided α=5%, power=80%, 10% follow-up allowance; not profit-powered.
Harm stratum: low usage AND zero overage AND zero care; missing ≠ zero. Looks at 50%/100%; one-sided Fisher p≤0.005 each plus positive churn difference (familywise ≤0.01).

References | Market and comparable implementations

[S1] T-Mobile US. 2025 Annual Report / Form 10-K (published 2026). Churn and ARPU definitions.
https://s29.q4cdn.com/310188824/files/doc_financials/2025/ar/2025-TMUS-Annual-Report-Final-Web.pdf
FY2025 monthly churn: postpaid phone 0.93%, prepaid 2.72%. Benchmarks do not set Company A's baseline.

[S2] AT&T. 2025 Form 10-K (9 February 2026). Mobility statistics and device-financing discussion.
https://www.sec.gov/Archives/edgar/data/732717/000073271726000120/t-20251231.htm
Operator churn definitions differ; device-financing maturity is not the same as handset age.

[S3] Verizon. Third-quarter 2026 results (26 October 2026). Retention cost and subsidy discipline.
https://www.verizon.com/about/news/verizon-delivers-another-record-breaking-quarter
30% promotional-retention-cost improvement is management-reported, not an experimental treatment effect.

[S7] MCE Systems. Vodafone Turkey launches Advanced Device Care by MCE and Techpo (26 May 2020).
https://www.mce.systems/articles/vodafone-turkey-launches-advanced-device-care-by-mce-and-techpo
Named diagnostics deployment in >1,000 stores. Vendor case supports feasibility, not quantified causal churn impact.

[S8] PegaWorld iNspire (2023). Vodafone's value-driven retention experience in assisted and digital channels.
https://www.pega.com/insights/resources/pegaworld-inspire-2023-how-vodafone-uses-pega-deliver-value-driven-retention
Operator/vendor workflow description. No reported percentage gain is transferred into Company A's economics.

References | Methods, data and interpretation

[S4] Ascarza, Iyengar & Schleicher (2016). Plan Recommendations; JMR 53(1), 46–60.
Randomized intervention: 6% control vs 10% treated three-month churn—not a universal service-contact effect.

[S5] Ascarza (2018). Retention Futility; JMR 55(1), 80–98.
Risk and treatment responsiveness differ; Company A has no treatment labels.

[S6] Lemmens & Gupta (2020). Managing Churn to Maximize Profits; Marketing Science 39(5), 956–973.
Incremental cash flow after all intervention costs—not risk scores alone—is the commercial objective.

[M1] scikit-learn: evaluation and calibration.

[M2] LightGBM: local CPU boosting.

[M5] Saerens et al. (2002): prior adjustment.

[M6] Elkan (2001): costs / known-prior correction.

[M3] XGBoost; [M4] CatBoost: comparators.

[M7] Deng et al. (2013): pre-period adjustment.

[M8] Neslin et al. (2006): historical churn protocol.

No matched AUC ceiling or measured CUPED claimed.

[D1–D2] Supplied data, overview, assignment and tutorial; all analytical figures are original local outputs.
Reproduction: philopateer.ipynb; frozen split, seed, packages and input hashes. Full URLs and evidence-scope caveats: references.txt.